s9b_inverse_vol

equity strategy | family: equity-volatility | status: validated | derived from `s6_neutral_levered`

What this strategy does

Strategy 6 weights x 1/ivol_252d within each neutralization cell

Why it should work. PEAD is a stronger, cleaner signal in low-vol stocks; size positions by inverse risk.

Design notes (from the strategy's source file).

Strategy 9b -- Strategy 6 sized by inverse idiosyncratic volatility.

Within every (quarter, side, sector, size) cell, each event's weight is scaled by 1/ivol, so calm stocks get more capital than wild ones. Rationale: the earnings drift is worth more per unit of risk in low-vol names (the SUE x vol interaction is t = -2.4 to -2.7 in Fama-MacBeth), and the engine's cost/liquidity model charges the same bps regardless of vol. Raises Sharpe 1.30 -> 1.47 with no fitted parameter, in all three sub-eras.

`InverseVolVolMatched` re-sizes it to Strategy 6's exact volatility to answer "at the SAME risk, how much more return": 7.31% vs 6.49%.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

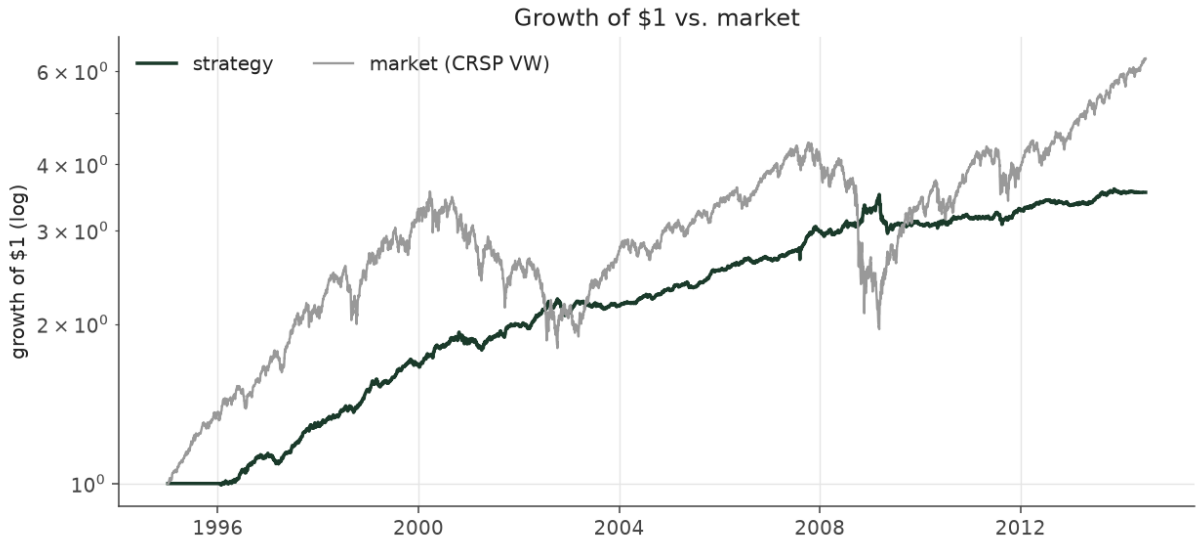
metric	s9b inverse vol	market (CRSP VW)
Annualized return	6.71%	9.94%
Annualized volatility	4.49%	19.35%
Sharpe (NAV-pct)	1.47	0.59
Sortino	2.17	
Max drawdown	-15.6%	-55.5%
Calmar	0.43	
Final NAV ($10M start)	$35,421,134	
Worst day	-3.36%	
% positive days	52.4%	
Sharpe 1996-2001	2.40	
Sharpe 2002-2007	1.63	
Sharpe 2008-2013	0.59	

Correlation of daily returns with the market: **0.01**.

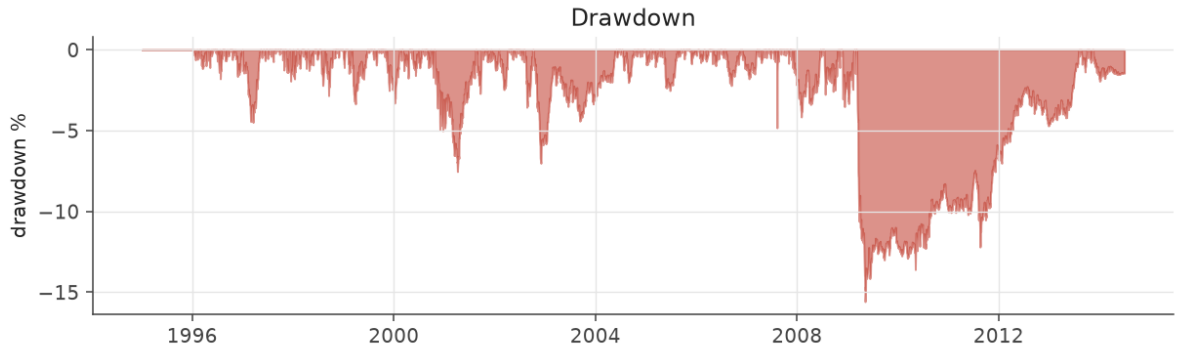
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 1.47

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.0671

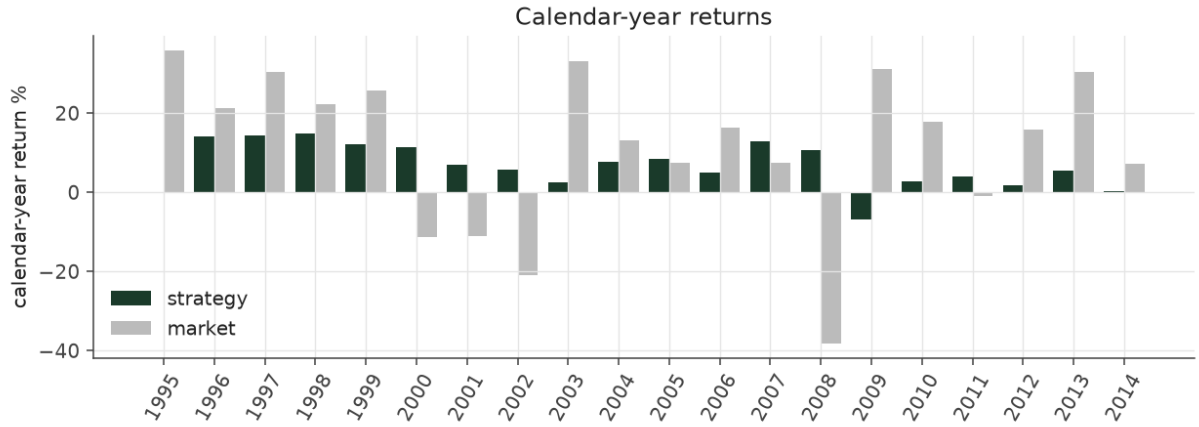
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.43



NAV (log scale)



Drawdown from running peak

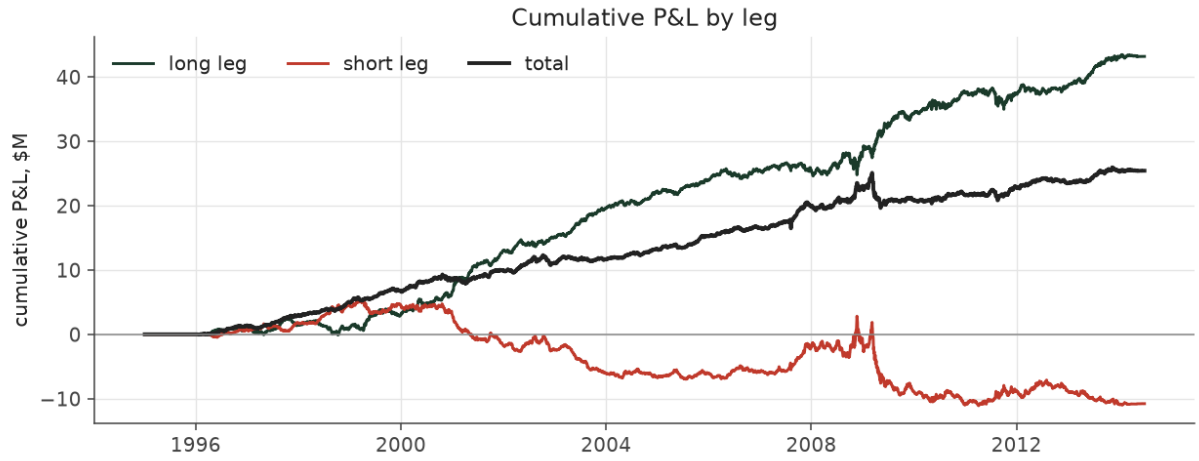


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	43.18	-10.78
P&L after costs ($M)	39.82	-14.40
of which common (average-event) component ($M)	27.28	-27.98
of which excess / PEAD component ($M)	15.90	17.20
excess minus costs ($M)	12.54	13.57
gross return per $ traded (bp)	253.23	-61.58
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.33	-0.23

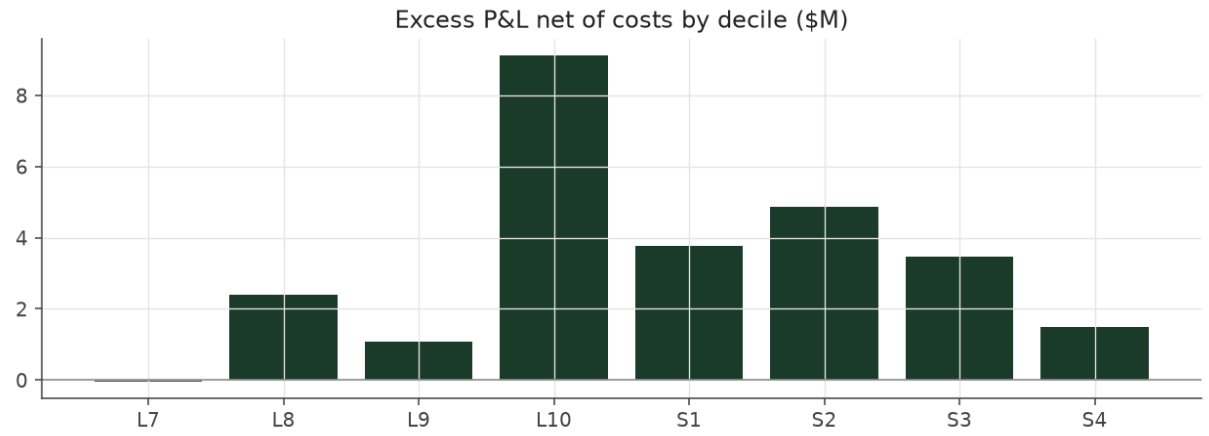
Daily P&L correlation between the two legs: **-0.81**.

• **The short leg loses money in absolute terms** (-10.8M before costs) while the long leg earns +43.2M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 27.3M / 28.0M. Stripping that common component, the surprise signal itself earned +15.9M on the long side and +17.2M on the short side.

• The two legs' daily P&L are -0.81 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.33 long / -0.23 short vs 1.47 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	151.20	2.79	0.25	0.30	-0.05	-0.2%	-3.44
long	8	16,588	401.72	9.02	3.19	0.80	2.39	9.1%	59.48
long	9	17,747	528.00	8.81	2.11	1.02	1.08	4.2%	20.55
long	10	17,174	624.27	19.20	10.36	1.24	9.12	34.9%	146.09
short	1	17,169	625.49	-5.50	5.11	1.35	3.76	14.4%	60.15
short	2	17,281	542.85	-4.32	5.97	1.11	4.86	18.6%	89.51
short	3	15,970	414.21	-3.31	4.31	0.84	3.47	13.3%	83.83
short	4	9,171	168.10	-1.27	1.80	0.32	1.48	5.7%	88.07



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	299.80	5.47	7.38	1.16	6.22	23.8%	207.43
2	22,885	638.69	5.81	7.99	1.82	6.17	23.6%	96.63
3	25,501	791.87	6.65	8.18	1.68	6.50	24.9%	82.09
4	27,583	858.14	4.41	5.65	1.36	4.28	16.4%	49.90
5	28,947	867.35	3.09	3.90	0.96	2.94	11.3%	33.93

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	228.45	5.04	6.62	1.14	5.47	21.0%	239.60
2	23,084	704.48	5.79	8.86	2.11	6.75	25.8%	95.75
3	25,851	863.36	6.82	8.95	1.73	7.22	27.7%	83.66
4	27,665	841.45	4.75	5.25	1.18	4.07	15.6%	48.33

5	28,874	817.81	3.00	3.39	0.82	2.57	9.8%	31.43

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	1,327.77	0.88	8.30	2.36	5.94	22.7%	44.73
41-60d	25,652	790.84	4.93	5.08	1.62	3.46	13.2%	43.73
61-90d	35,324	995.63	7.27	8.77	2.12	6.64	25.4%	66.73
>90d	12,298	341.59	12.34	10.95	0.87	10.07	38.6%	294.91

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	689.80	9.68	11.33	1.37	9.96	38.1%	144.39
2002-07	39,956	1,152.94	10.10	12.32	2.28	10.04	38.5%	87.10
2008-13	45,278	1,613.11	5.64	9.45	3.33	6.11	23.4%	37.90

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	341.69	2.19	2.92	0.73	2.19	8.4%	63.98
Chems	2,590	229.95	1.46	1.85	0.40	1.45	5.5%	62.94
Durbl	2,843	248.11	2.35	2.86	0.45	2.41	9.2%	96.99
Enrgy	6,194	278.49	3.79	4.41	0.52	3.89	14.9%	139.59
Hlth	11,696	330.66	2.61	3.32	0.70	2.62	10.0%	79.16
Manuf	12,034	309.06	1.80	2.61	0.65	1.97	7.5%	63.67
Money	17,583	302.38	2.62	3.41	0.69	2.72	10.4%	90.01
NoDur	5,081	278.21	1.44	1.95	0.53	1.42	5.4%	50.88
Other	18,377	329.38	2.59	3.42	0.74	2.68	10.2%	81.25
Shops	11,984	320.79	3.15	3.83	0.67	3.16	12.1%	98.56
Telcm	3,981	270.71	0.85	1.39	0.49	0.89	3.4%	33.01
Utils	4,672	216.42	0.56	1.13	0.40	0.73	2.8%	33.97

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	847.40	4.09	6.58	1.56	5.01	19.2%	59.15
low vol	40,091	1,438.79	11.70	15.05	3.10	11.95	45.8%	83.08
mid vol	37,903	1,120.54	9.45	11.15	2.21	8.94	34.2%	79.76

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	6.71%	0.00%
0.5%	1.8	6.43%	0.28%
1.0%	3.5	6.13%	0.57%
2.0%	7.1	5.50%	1.21%
5.0%	17.7	2.99%	3.72%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.